

INTERNOVA

WEEK 1 ASSIGNMENT

Python Fundamentals for Data Analytics

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Branch: Computer Science and Engineering

Offer Letter ID: INOL/26/DA/633

Week: 1

Task 1 — Python Basics

Objective:

To understand basic Python input, output, and formatted printing.

Python Source Code:

```
print("Welcome to Python Data Analytics!")
name = input("Enter your name: ")
college = input("Enter your college name: ")
branch = input("Enter your branch: ")
print("\n----- Student Information -----")
print("Name   :", name)
print("College:", college)
print("Branch :", branch)
print("-----")
```

Output:

```
... Welcome to Python Data Analytics
   Enter Your Name:Saga Tejaswi Lakshmi Priya Durga
   Enter your college name:VKR & VNB Engineering College
   Enter your Branch:CSE

   ----- Student Information -----
   Name: Saga Tejaswi Lakshmi Priya Durga
   College: VKR & VNB Engineering College
   Branch: CSE
   -----
```

Explanation:

This task demonstrates basic Python programming by taking student details as input and displaying them in a formatted way.

Task 2 — Variables & Data Types

Objective: To understand Python variables and different data types such as integer, float, string, and Boolean.

Python Source Code:

```
age = 20
cgpa = 8.26
name = "Tejaswi"
is_student = True

print("Age:", age)
print("Data Type:", type(age))

print("\nCGPA:", cgpa)
print("Data Type:", type(cgpa))
```

```
print("\nName:", name)
print("Data Type:", type(name))

print("\nStudent:", is_student)
print("Data Type:", type(is_student))
```

Output:

```
... Age: 20
    Data Type: <class 'int'>

    CGPA: 8.26
    Data Type: <class 'float'>

    Name: Saga Tejaswi
    Data Type: <class 'str'>

    Student: True
    Data Type: <class 'bool'>
```

Explanation:

This task demonstrates different Python data types such as integer, float, string, and Boolean using variables and the type() function.

Task 3: Operators

Objective: To understand and perform basic arithmetic operations such as addition, subtraction, multiplication, division, and modulus.

Python Source Code:

```
num1 = float(input("Enter first number: "))
num2 = float(input("Enter second number: "))
print("\n----- Calculator -----")
print("Addition:", num1 + num2)
print("Subtraction:", num1 - num2)
print("Multiplication:", num1 * num2)
if num2 != 0:
    print("Division:", num1 / num2)
    print("Modulus:", num1 % num2)
else:
    print("Division: Cannot divide by zero")
    print("Modulus: Cannot divide by zero")
```

Output:

```
... Enter first number: 10
    Enter second number: 3

    ----- Calculator -----
    Addition: 13.0
    Subtraction: 7.0
    Multiplication: 30.0
    Division: 3.3333333333333335
    Modulus: 1.0
```

Explanation: This task creates a simple calculator using two numbers and performs addition, subtraction, multiplication, division, and modulus operations.

Task 4: Conditional Statements

Objective: To learn how to use if, elif, and else statements to make decisions based on given conditions.

Python Source Code:

```
marks = float(input("Enter your marks: "))
if marks >= 90:
    grade = "A"
elif marks >= 75:
    grade = "B"
elif marks >= 60:
    grade = "C"
else:
    grade = "Fail"
print("Your Grade is:", grade)
```

Output:

```
... Enter your marks: 61
    Your Grade is: C
```

Explanation: This task uses if, elif, and else statements to calculate and display a grade based on the marks entered by the user.

Task 5: Loops

Objective: To understand and use for and while loops for repeating tasks and processing a sequence of values.

Python Source Code:

```
print("Numbers from 1 to 20:")
for i in range(1, 21):
    print(i)
```

```
num = int(input("Enter a number: "))
print("\nMultiplication Table")
for i in range(1, 11):
    print(num, "x", i, "=", num * i)
```

```
i = 2
print("\nEven numbers from 1 to 50:")
while i <= 50:
    print(i)
    i += 2
```

Output:

Numbers from 1 to 20:

1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20

Enter a number: 15

Multiplication Table

15 x 1 = 15
15 x 2 = 30
15 x 3 = 45
15 x 4 = 60
15 x 5 = 75
15 x 6 = 90
15 x 7 = 105
15 x 8 = 120
15 x 9 = 135
15 x 10 = 150

Even numbers from 1 to 50:

2
4
6
8
10
12
14
16
18
20
22
24
26
28
30
32
34
36
38
40
42
44
46
48
50

Explanation: This task demonstrates for and while loops by printing numbers, generating a multiplication table, and displaying even numbers.

Task 6: Functions

Objective: To learn how to create and use user-defined functions to perform specific calculations and make programs reusable.

Python Source Code:

```
def calculate_square(number):  
    return number * number  
def calculate_average(a, b, c):  
    return (a + b + c) / 3  
num = float(input("Enter a number: "))  
print("Square:", calculate_square(num))
```

```

print("\nEnter three numbers:")
a = float(input("First number: "))
b = float(input("Second number: "))
c = float(input("Third number: "))
print("Average:", calculate_average(a, b, c))

```

Output:

```

Enter a number: 61
Square: 3721.0

```

```

Enter three numbers:
First number: 5
Second number: 6
Third number: 3
Average: 4.666666666666667

```

Explanation: This task demonstrates user-defined functions to calculate the square of a number and the average of three numbers.

Task 7: Strings & Collections

Objective: To understand string operations and use Python collections such as lists, tuples, dictionaries, and sets for storing and managing data.

Python Source Code:

```

#===== String =====
text = "Python Data Analytics"
print("Original:", text)
print("Uppercase:", text.upper())
print("Lowercase:", text.lower())
print("Replace:", text.replace("Python", "Excel"))
print("Position of Data:", text.find("Data"))
#===== List =====
numbers = [10, 20, 30, 40]
print("\nOriginal list:", numbers)
numbers.append(50)
print("After append:", numbers)
numbers.remove(20)
print("After remove:", numbers)
numbers.sort()
print("After sort:", numbers)
#===== Tuple =====
subjects = ("Python", "SQL", "Power BI", "Excel")
print("\nTuple:", subjects)
print("First subject:", subjects[0])
print("Second subject:", subjects[1])
#===== Dictionary =====
student = {
    "Name": "Tejaswi",

```

```

"College": "JNTUK",
"Branch": "CSE",
"CGPA": 8.26
}
print("\nStudent Information:")
print(student)
print("Name:", student["Name"])
print("College:", student["College"])
print("Branch:", student["Branch"])
print("CGPA:", student["CGPA"])
skills = {"Python", "SQL", "Excel"}
print("Original set:", skills)
#===== Set =====
skills.add("Power BI")
print("\nAfter add:", skills)
skills.remove("Excel")
print("After remove:", skills)

```

Output:

```

Original: Python Data Analytics
Uppercase: PYTHON DATA ANALYTICS
Lowercase: python data analytics
Replace: Excel Data Analytics
Position of Data: 7

```

```

Original list: [10, 20, 30, 40]
After append: [10, 20, 30, 40, 50]
After remove: [10, 30, 40, 50]
After sort: [10, 30, 40, 50]

```

```

Tuple: ('Python', 'SQL', 'Power BI', 'Excel')
First subject: Python
Second subject: SQL

```

```

Student Information:
{'Name': 'Tejaswi', 'College': 'JNTUK', 'Branch': 'CSE', 'CGPA': 8.26}
Name: Tejaswi
College: JNTUK
Branch: CSE
CGPA: 8.26
Original set: {'Excel', 'SQL', 'Python'}

```

```

After add: {'Excel', 'Power BI', 'SQL', 'Python'}
After remove: {'Power BI', 'SQL', 'Python'}

```

Explanation: This task demonstrates string operations and Python collections such as lists, tuples, dictionaries, and sets.

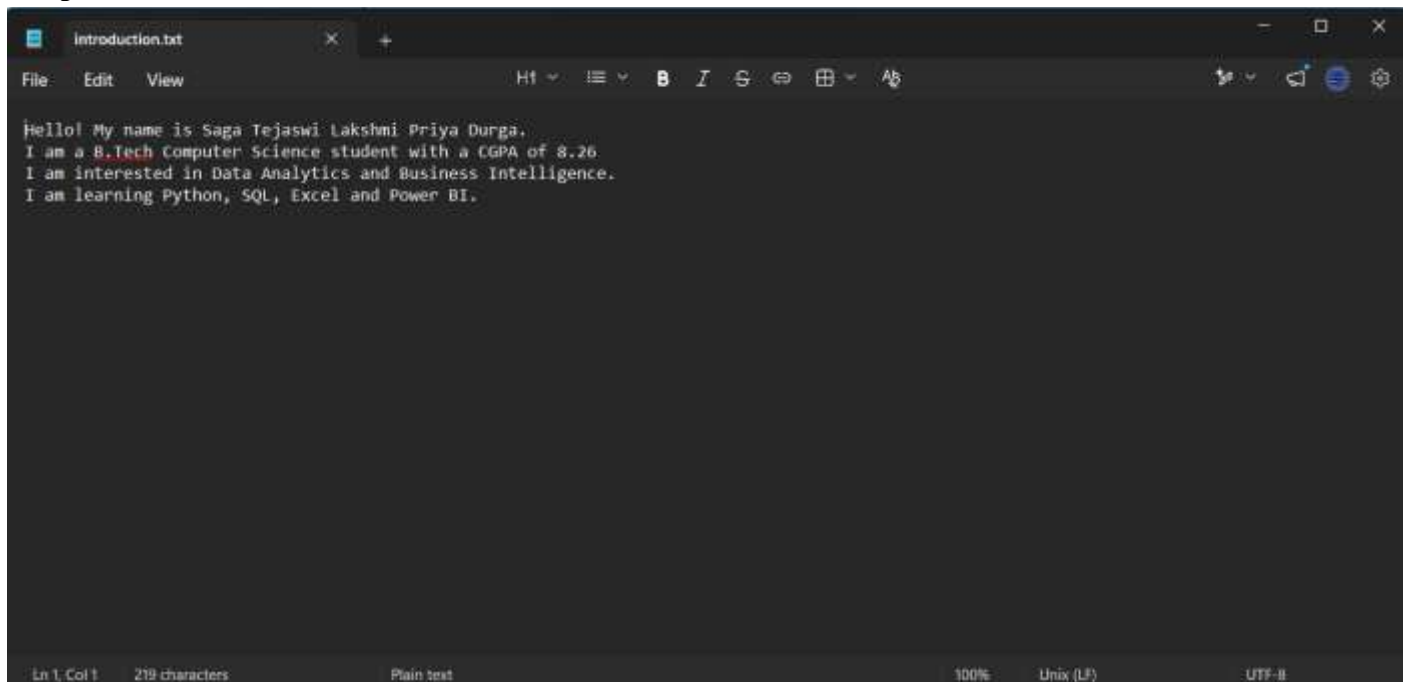
Task 8: Basic File Handling

Objective: To learn how to create, write, read, and display information from a text file using Python.

Python Source Code:

```
filename = "introduction.txt"
# Create and write to the file
with open(filename, "w") as file:
    file.write("Hello! My name is Tejaswi.\n")
    file.write("I am a B.Tech Computer Science student.\n")
    file.write("I am interested in Data Analytics and Business Intelligence.\n")
    file.write("I am learning Python, SQL, Excel and Power BI.\n")
# Read and display the file
with open(filename, "r") as file:
    content = file.read()
print("----- File Contents -----")
print(content)
```

Output:

A screenshot of a text editor window titled 'introduction.txt'. The window has a dark theme and a menu bar with 'File', 'Edit', and 'View'. The toolbar includes icons for font size, bold, italic, underline, and other text formatting options. The text area contains the following content: 'Hello! My name is Saga Tejaswi Lakshmi Priya Durga.', 'I am a B.Tech Computer Science student with a CGPA of 8.26', 'I am interested in Data Analytics and Business Intelligence.', and 'I am learning Python, SQL, Excel and Power BI.'. The status bar at the bottom shows 'Ln 1, Col 1', '219 characters', 'Plain text', '100%', 'Unix (LF)', and 'UTF-8'.

----- File Contents -----

```
Hello! My name is Saga Tejaswi Lakshmi Priya Durga.
I am a B.Tech Computer Science student with a CGPA of 8.26
I am interested in Data Analytics and Business Intelligence.
I am learning Python, SQL, Excel and Power BI.
```

Explanation: This task demonstrates file handling by creating a text file, writing an introduction into it, and reading and displaying its contents.

Task 9: Student Record Management System

Objective: To develop a simple Python-based student record management system that can add, display, search, and delete student records using lists or dictionaries.

Python Source Code:

```
# Student Record Management System
students = []
def add_student():
    print("\n--- Add Student ---")
    name = input("Enter student name: ")
```

```

roll_no = input("Enter roll number: ")
branch = input("Enter branch: ")
marks = float(input("Enter marks: "))
student = {
    "Name": name,
    "Roll No": roll_no,
    "Branch": branch,
    "Marks": marks
}
students.append(student)
print("Student record added successfully!")
def display_students():
    print("\n--- All Student Records ---")
    if len(students) == 0:
        print("No student records found.")
        return
    for student in students:
        print("Name :", student["Name"])
        print("Roll No:", student["Roll No"])
        print("Branch :", student["Branch"])
        print("Marks :", student["Marks"])
        print("-----")
def search_student():
    print("\n--- Search Student ---")
    name = input("Enter student name to search: ")
    found = False
    for student in students:
        if student["Name"].lower() == name.lower():
            print("\nStudent Found!")
            print("Name :", student["Name"])
            print("Roll No:", student["Roll No"])
            print("Branch :", student["Branch"])
            print("Marks :", student["Marks"])
            found = True
    if not found:
        print("Student not found.")
def delete_student():
    print("\n--- Delete Student ---")
    roll_no = input("Enter roll number to delete: ")
    for student in students:
        if student["Roll No"] == roll_no:
            students.remove(student)
            print("Student record deleted successfully!")
            return
    print("Student record not found.")
while True:
    print("\n===== Student Record Management System =====")
    print("1. Add Student")
    print("2. Display All Students")
    print("3. Search Student by Name")
    print("4. Delete Student")
    print("5. Exit")
    choice = input("Enter your choice: ")
    if choice == "1":
        add_student()

```

```
elif choice == "2":
    display_students()
elif choice == "3":
    search_student()
elif choice == "4":
    delete_student()
elif choice == "5":
    print("Thank you for using the system!")
    break
else:
    print("Invalid choice. Please try again.")
```

Output:

```
>>> %Run -c $EDITOR_CONTENT
```

```
===== Student Record Management System =====
```

1. Add Student
2. Display All Students
3. Search Student by Name
4. Delete Student
5. Exit

```
Enter your choice: 1
```

```
--- Add Student ---
```

```
Enter student name: S.Tejaswi
```

```
Enter roll number: 1
```

```
Enter branch: CSE
```

```
Enter marks: 86
```

```
Student record added successfully!
```

```
===== Student Record Management System =====
```

1. Add Student
2. Display All Students
3. Search Student by Name
4. Delete Student
5. Exit

Enter your choice: 1

--- Add Student ---

Enter student name: P.Jyothika

Enter roll number: 2

Enter branch: CSE

Enter marks: 84

Student record added successfully!

===== Student Record Management System =====

1. Add Student

2. Display All Students

3. Search Student by Name

4. Delete Student

5. Exit

Enter your choice: 3

--- Search Student ---

Enter student name to search: T.Vasanth

Student not found.

===== Student Record Management System =====

1. Add Student

2. Display All Students

3. Search Student by Name

4. Delete Student

5. Exit

Enter your choice: 2

--- All Student Records ---

Name : S.Tejaswi

Roll No: 1

Branch : CSE

Marks : 86.0

Name : P.Jyothika

Roll No: 2

Branch : CSE

Marks : 84.0

===== Student Record Management System =====

1. Add Student

2. Display All Students

3. Search Student by Name

4. Delete Student

5. Exit

Enter your choice: 2

--- All Student Records ---

Name : S.Tejaswi

Roll No: 1

Branch : CSE

Marks : 86.0

Name : P.Jyothika

Roll No: 2

Branch : CSE

Marks : 84.0

==== Student Record Management System ====

1. Add Student

2. Display All Students

3. Search Student by Name

4. Delete Student

5. Exit

Enter your choice:

Explanation:

This mini project uses a list of dictionaries to store student records and provides options to add, display, search, and delete student information.